

Comparison of Trees for Graduation Prediction: UMBC vs. USM

Comparison of UMBC and USM Trees (Decision Tree)

1. Root Node Split

- **UMBC:** `totalsupport <= 31429.50`
- **USM:** `totalsupport <= 11177.50`

Observation: Both trees use `totalsupport` as the root feature, indicating its primary importance in both datasets. UMBC uses a higher threshold, possibly reflecting a higher range of financial support in that dataset.

2. Primary Branch Structures

UMBC Tree – Left Subtree (`totalsupport <= 31429.50`)

- Next split: `supportedY1`
- If `supportedY1 <= 0.50`:
 - Split on `Changed_Major_No`
 - If `<= 0.50`:
 - * Further split on `Got_Merit_No` → All paths: **class 1**
 - If `> 0.50`:
 - * Split on `totalsupport <= 87.50`
 - * `<= 87.50`: **class 0**
 - * `> 87.50`: **class 1**
- If `supportedY1 > 0.50`:

- Split on `totalsupport <= 15936.00`
- Followed by `Changed_Major_No` → All outcomes: **class 0**
- If `totalsupport > 15936.00`:
 - * `<= 20140.00`: **class 1**
 - * `> 20140.00`: **class 0**

USM Tree – Left Subtree (`totalsupport <= 11177.50`)

- Next split: `Changed_Major_No`
- If `<= 0.50`:
 - Split on `EFC_Category_High`
 - If `totalsupport <= 3579.00`: **class 0**, else **class 1**
 - If `EFC_Category_High > 0.50`: all branches → **class 0**
- If `> 0.50`:
 - Split on `EFC_Category_High`
 - If `<= 0.50`:
 - * Split on `Got_Merit_Yes` → Both branches: **class 1**
 - If `> 0.50`:
 - * `totalsupport <= 4490.50`: **class 0**, else **class 1**

Observation: UMBC’s left subtree emphasizes `supportedY1` and `Got_Merit_No`, showing nuanced support patterns. USM focuses more on `EFC_Category_High` and `Changed_Major_No`, using `totalsupport` only after those.

3. Right Subtree (Higher Support Students)

UMBC Tree – Right Subtree (`totalsupport > 31429.50`)

- Split on `Got_Scholar_Yes`
- If `<= 0.50`:
 - `totalsupport <= 124977.50` → **class 1**
 - `> 124977.50`:
 - * `<= 142881.00`: **class 0**
 - * `> 142881.00`: **class 1**
- If `> 0.50`:

- `totalsupport <= 59605.00` → **class 1**
- `> 59605.00`:
 - * `<= 122406.00`: **class 1**
 - * `> 122406.00`: **class 0**

USM Tree – Right Subtree (`totalsupport > 11177.50`)

- Split on `totalsupport <= 25156.00`
- If `EFC_Category_High <= 0.50`:
 - Split on `totalsupport <= 13911.00` → **class 1**
- If `EFC_Category_High > 0.50`:
 - Split on `Changed_Major_No` → All: **class 1**
- If `totalsupport > 25156.00`:
 - Split at 39531.50 → All: **class 1**

Observation: UMBC uses fine-grained splits with scholarship flags and large support thresholds. USM simplifies with broad `totalsupport` cutoffs and emphasizes EFC but almost all paths → **class 1**.

4. Summary of Branch Outcomes

This section explores what kinds of students were predicted to graduate (`class: 1`) or not graduate (`class: 0`), based on how the decision branches unfold in each tree.

UMBC Tree

Aspect	Description
Class 0 Predictions	Yes, there are specific conditions under which students are predicted not to graduate. These occur primarily under: • Low <code>totalsupport</code> (<code><= 31429.50</code>) • No early support (<code>supportedY1 <= 0.50</code>) • <code>Changed_Major_No > 0.5</code> • Very low aid (<code>totalsupport <= 87.50</code>) These precise combinations of financial and academic behaviors lead to a <code>class: 0</code> (not graduated) outcome.
Class 1 Predictions	A majority of the branches lead to a graduated (<code>class: 1</code>) outcome. For example: • Moderate or high <code>totalsupport</code> with <code>Got_Scholar_Yes = 1</code> • Early support (<code>supportedY1 > 0.5</code>) and <code>totalsupport</code> in a midrange (15936–20140) • Even in cases with <code>Got_Merit_No = 1</code>, graduation is predicted if <code>totalsupport</code> is sufficiently high
Outcome Balance	The model distinguishes clearly between students who are likely to graduate vs. not. There are several paths to both <code>class 0</code> and <code>class 1</code>, based on nuanced combinations of support timing, aid type, and academic decisions.
Decision Depth	Deeper decision paths with fine-grained thresholds (e.g., 87.50, 39906.50) enable granular modeling of edge cases.

USM Tree

Aspect	Description
Class 0 Predictions	<code>class: 0</code> (not graduated) outcomes are rare and occur only under very specific low-support and high-EFC conditions, such as: • <code>totalsupport <= 3579.00</code> and <code>Changed_Major_No <= 0.5</code> • <code>EFC_Category_High > 0.5</code> and <code>totalsupport <= 4899.00</code> These reflect students from higher-income families who received little to no aid and did not change majors.
Class 1 Predictions	Most branches — even under moderate support levels — lead to graduated (<code>class: 1</code>) predictions. For instance: • All branches where <code>totalsupport > 11177.50</code> lead to <code>class 1</code>, regardless of <code>EFC_Category_High</code> or <code>Changed_Major_No</code> • Even students with <code>Changed_Major_No > 0.5</code> are predicted to graduate if <code>totalsupport</code> exceeds a certain threshold
Outcome Balance	The model is strongly skewed toward predicting graduation, especially for students with support above \$11,000 or those who made an academic shift.
Decision Depth	Shallower and broader decision structure — few nodes go beyond 3 levels deep. This indicates a simpler model that focuses on major indicators like <code>EFC</code> and <code>totalsupport</code> .

5. Final Feature Coverage

Feature	UMBC Tree	USM Tree
<code>totalsupport</code>	Present	Present
<code>Changed_Major_No</code>	Present	Present
<code>Got_Merit</code>	Present (as No flag)	Present (as Yes flag)
<code>Got_Scholar_Yes</code>	Present	Absent
<code>supportedY1</code>	Present	Absent
<code>EFC_Category_High</code>	Absent	Present

6. Key Takeaways

- Both decision trees prioritize `totalsupport`, though they apply different thresholds and interpret its effect within different contextual conditions.
- The UMBC tree incorporates indicators of timing (`supportedY1`) and scholarship type (`Got_Scholar_Yes`), resulting in a more nuanced decision structure.
- The USM tree emphasizes financial background through `EFC_Category_High`, with simpler branching logic based on broader ranges of support.
- In the UMBC model, both `class 0` (not graduated) and `class 1` (graduated) outcomes are predicted under different combinations of variables, reflecting a more balanced outcome distribution.
- The USM model predominantly predicts `class 1` outcomes, especially for students receiving financial support above a certain threshold, indicating stronger influence of financial aid level.

Comparison of UMBC and USM Trees (XGBoost)

7. Root Split Comparison Table

Dataset	Root Condition	Interpretation
UMBC	<code>totalsupport <= 31429.50</code>	Uses a higher financial support threshold as the first decision node
USM	<code>totalsupport <= 11177.50</code>	Starts from a lower support threshold, indicating lower aid distribution

Observation: Both trees prioritize `totalsupport` at the root level, highlighting its critical role in predicting graduation.

8. Left Subtree Structure (Low Support Students)

UMBC Tree (`totalsupport <= 31429.50`)

- Splits on `supportedY1` (early year aid)
- Then splits on:
 - `Changed_Major_No`
 - `Got_Merit_No`
- Final micro-split on `totalsupport <= 87.50`

Outcome:

- Most paths lead to `class: 1` (graduated)
- But very low support (87.50) + no early support + no major change → `class: 0`

USM Tree (11177.50)

- Splits on `Changed_Major_No`
- Then branches on:
 - `EFC_Category_High`
 - `totalsupport` (again, for fine-grained cutoff)
 - `Got_Merit_Yes`

Outcome:

- Specific paths with high EFC + low support predict `class: 0`
- Other low support paths lead to `class: 1`, especially if there was a major change

9. Right Subtree Structure (High Support Students)

UMBC Tree (totalsupport > 31429.50)

- Splits on `Got_Scholar_Yes`
- Then branches through large `totalsupport` thresholds (e.g., 124977.50, 142881.00)

Outcome:

- Vast majority of high-support students → `class: 1`
- Exception: extremely high support (> 142881.00) **without scholarship** → `class: 0`

USM Tree (totalsupport > 11177.50)

- Splits on `totalsupport <= 25156.00` or higher
- Then considers `EFC_Category_High` and `Changed_Major_No`

Outcome:

- All branches on this side of the tree lead to `class: 1`
- High support almost guarantees graduation

10. Summary of Branch Outcomes

Aspect	UMBC Tree	USM Tree
Class 0 predictions	Present in fine-grained paths (e.g., very low support + no merit + no early aid)	Limited to very low support + high EFC cases
Class 1 predictions	Found across most branches; merit/scholarship aid reinforces	Dominates the tree beyond support $\geq$ 11k
Support Thresholds	Used with more precision and smaller margins	Broader buckets (e.g., 11k, 25k, 39k)
Aid Timing Used?	Yes (<code>supportedY1</code>)	No
Scholarship Indicator Used?	Yes (<code>Got_Scholar_Yes</code>)	No
EFC Category Used?	No	Yes

11. Conclusion

The UMBC Tree offers a granular decision-making model, using multiple financial and academic features—such as early aid support and scholarship indicators—with fine-grained splits to capture nuanced student outcomes.

In contrast, the USM Tree uses a simpler structure, relying more heavily on broad `totalsupport` thresholds and the `EFC_Category_High` feature. This leads to a strong skew toward predicting graduation for any student who receives moderate support, emphasizing general trends over individual nuance.